

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

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## Device Specifications

Device Type: Video Projector  
Manufacturer: Sharp  
Firmware Version: N/A  
Model(s): XG-C335X, XG-C430X, XG-C330X, XG-C435X, XG-C465X

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 2.06.0002-b002                         | 1.4.2                   |

## Version History

| Module Version | Date      | Notes           |
|----------------|-----------|-----------------|
| 1_0_0_0        | 1/19/2018 | Initial Version |

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## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`

## Supported Classes and Examples

|                                                                                                            |
|------------------------------------------------------------------------------------------------------------|
| <b>SerialClass</b>                                                                                         |
| <code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='XG-C335X')</code>               |
| <b>SerialOverEthernetClass</b>                                                                             |
| <code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='XG-C335X')</code> |

## Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                                         |                                                       |                                           |                                          |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------|------------------------------------------|
| Command<br>AspectRatio                                                                  | Value<br>'Normal'<br><br>'Stretch'<br><br>'V-Stretch' | Value<br>'Full'<br><br>'Border'           | Value<br>'Dot by Dot'<br><br>'Area Zoom' |
| Qualifier Key<br>'Input'                                                                | Qualifier Value<br>'RGB 1'<br><br>'S-Video'           | Qualifier Value<br>'RGB 2'<br><br>'Video' | Qualifier Value<br>'DVI'                 |
| # AspectRatio example<br>InterfaceName.Set('AspectRatio', 'Normal', {'Input': 'RGB 1'}) |                                                       |                                           |                                          |
| Command<br>AutoImage                                                                    | Value<br>None                                         |                                           |                                          |
| # AutoImage example<br>InterfaceName.Set('AutoImage', None)                             |                                                       |                                           |                                          |
| Command<br>AVMute                                                                       | Value<br>'On'                                         | Value<br>'Off'                            |                                          |
| # AVMute example<br>InterfaceName.Set('AVMute', 'On')                                   |                                                       |                                           |                                          |
| Command<br>Freeze                                                                       | Value<br>'On'                                         | Value<br>'Off'                            |                                          |
| # Freeze example<br>InterfaceName.Set('Freeze', 'On')                                   |                                                       |                                           |                                          |
| Command<br>Input                                                                        | Value<br>'RGB 1'<br><br>'S-Video'                     | Value<br>'RGB 2'<br><br>'Video'           | Value<br>'DVI'                           |
| # Input example<br>InterfaceName.Set('Input', 'RGB 1')                                  |                                                       |                                           |                                          |
| Command<br>LampMode                                                                     | Value<br>'Eco'                                        | Value<br>'Normal'                         |                                          |
| # LampMode example<br>InterfaceName.Set('LampMode', 'Eco')                              |                                                       |                                           |                                          |
| Command<br>Power                                                                        | Value<br>'On'                                         | Value<br>'Off'                            |                                          |
| # Power example<br>InterfaceName.Set('Power', 'On')                                     |                                                       |                                           |                                          |

## Global Scripter Module Communication Sheet

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| Command                                             | Value                 |
|-----------------------------------------------------|-----------------------|
| <b>Volume</b>                                       | 0 to 60 in steps of 1 |
| # Volume example<br>InterfaceName.Set('Volume', 60) |                       |

## Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

### Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

### Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                                                                 |                                                                     |                                                                    |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------|
| <b>Command</b><br><b>AspectRatio</b>                                                                                                                                                                                            | Value<br>'Normal'<br>'Stretch'<br>'V-Stretch'                       | Value<br>'Full'<br>'Border'                                        | Value<br>'Dot by Dot'<br>'Area Zoom'   |
| <b>Qualifier Key</b><br><b>'Input'</b>                                                                                                                                                                                          | Qualifier Value<br>'RGB 1'<br>'S-Video'                             | Qualifier Value<br>'RGB 2'<br>'Video'                              | Qualifier Value<br>'DVI'               |
| # AspectRatio examples<br>InterfaceName.Update('AspectRatio', {'Input': 'RGB 1'})<br>Value = InterfaceName.ReadStatus('AspectRatio', {'Input': 'RGB 1'})<br>InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler) |                                                                     |                                                                    |                                        |
| <b>Command</b><br><b>AVMute</b>                                                                                                                                                                                                 | Value<br>'On'                                                       | Value<br>'Off'                                                     |                                        |
| # AVMute examples<br>InterfaceName.Update('AVMute')<br>Value = InterfaceName.ReadStatus('AVMute')<br>InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)                                                             |                                                                     |                                                                    |                                        |
| <b>Command</b><br><b>ConnectionStatus</b>                                                                                                                                                                                       | Value<br>'Connected'                                                | Value<br>'Disconnected'                                            |                                        |
| # ConnectionStatus examples<br>Value = InterfaceName.ReadStatus('ConnectionStatus')<br>InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)                                                                 |                                                                     |                                                                    |                                        |
| <b>Command</b><br><b>DeviceStatus</b>                                                                                                                                                                                           | Value<br>'Normal'<br>'Filter Cover Error'<br>'Temp Abnormally High' | Value<br>'Temp High'<br>'Lamp Burnt-out'<br>'Lamp Life 5% or Less' | Value<br>'Fan Error'<br>'Lamp Not Lit' |
| # DeviceStatus examples<br>InterfaceName.Update('DeviceStatus')                                                                                                                                                                 |                                                                     |                                                                    |                                        |

# Global Scripter Module Communication Sheet

Revision: 1/19/2018

|                                                                                                                                                                                     |                                |                                |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------|------------------|
| Value = InterfaceName.ReadStatus('DeviceStatus')<br>InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)                                                            |                                |                                |                  |
| <b>Command</b><br><b>Freeze</b>                                                                                                                                                     | Value<br>'On'                  | Value<br>'Off'                 |                  |
| # Freeze examples<br>InterfaceName.Update('Freeze')<br>Value = InterfaceName.ReadStatus('Freeze')<br>InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)                 |                                |                                |                  |
| <b>Command</b><br><b>Input</b>                                                                                                                                                      | Value<br>'RGB 1'<br>'S-Video'  | Value<br>'RGB 2'<br>'Video'    | Value<br>'DVI'   |
| # Input examples<br>InterfaceName.Update('Input')<br>Value = InterfaceName.ReadStatus('Input')<br>InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)                     |                                |                                |                  |
| <b>Command</b><br><b>LampMode</b>                                                                                                                                                   | Value<br>'Eco'                 | Value<br>'Normal'              |                  |
| # LampMode examples<br>InterfaceName.Update('LampMode')<br>Value = InterfaceName.ReadStatus('LampMode')<br>InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)         |                                |                                |                  |
| <b>Command</b><br><b>LampStatus</b>                                                                                                                                                 | Value<br>'On'<br>'Waiting'     | Value<br>'Off'<br>'Lamp Error' | Value<br>'Retry' |
| # LampStatus examples<br>InterfaceName.Update('LampStatus')<br>Value = InterfaceName.ReadStatus('LampStatus')<br>InterfaceName.SubscribeStatus('LampStatus', None, FeedbackHandler) |                                |                                |                  |
| <b>Command</b><br><b>LampUsage</b>                                                                                                                                                  | Value<br>0 –                   |                                |                  |
| # LampUsage examples<br>InterfaceName.Update('LampUsage')<br>Value = InterfaceName.ReadStatus('LampUsage')<br>InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)     |                                |                                |                  |
| <b>Command</b><br><b>Power</b>                                                                                                                                                      | Value<br>'On'                  | Value<br>'Off'                 |                  |
| # Power examples<br>InterfaceName.Update('Power')<br>Value = InterfaceName.ReadStatus('Power')<br>InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)                     |                                |                                |                  |
| <b>Command</b><br><b>Volume</b>                                                                                                                                                     | Value<br>0 to 60 in steps of 1 |                                |                  |
| # Volume examples<br>InterfaceName.Update('Volume')<br>Value = InterfaceName.ReadStatus('Volume')                                                                                   |                                |                                |                  |

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```
InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)
```

## Cable and Adapter Requirements

Captive Screw to 9 Pin MiniDIN RS-232 Serial Cable

## Notes for the Device

## Serial communication

Port Type: RS-232

Baud Rate: 9600

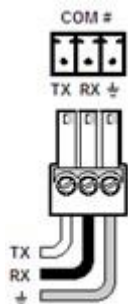
Data Bits: 8

Parity: None

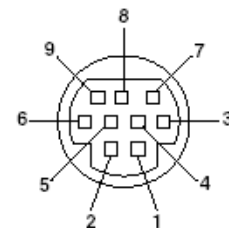
Stop Bits: One

Flow Control: None

## Pin Assignments Diagram



| Signal | Main Cable | Pin | Signal |
|--------|------------|-----|--------|
| RX     | ↗          | 2   | RD     |
| TX     | ↖          | 3   | TD     |
| GND    | —          | 5   | GND    |
| RTS    | ⌋          | 7   | RTS    |
| CTS    | ⌋          | 8   | CTS    |





## Appendix A. Set Commands

|                                       |             |
|---------------------------------------|-------------|
| AV Mute Off                           | IMBK 0\X0D  |
| AV Mute On                            | IMBK 1\X0D  |
| Aspect Ratio Area Zoom Input DVI      | RCSR 10\X0D |
| Aspect Ratio Border Input DVI         | RCSR 6\X0D  |
| Aspect Ratio Dot by Dot Input DVI     | RCSR 3\X0D  |
| Aspect Ratio Full Input DVI           | RCSR 5\X0D  |
| Aspect Ratio Normal Input DVI         | RCSR 1\X0D  |
| Aspect Ratio Stretch Input DVI        | RCSR 2\X0D  |
| Aspect Ratio V-Stretch Input DVI      | RCSR 11\X0D |
| Aspect Ratio Area Zoom Input RGB 1    | RASR 10\X0D |
| Aspect Ratio Border Input RGB 1       | RASR 6\X0D  |
| Aspect Ratio Dot by Dot Input RGB 1   | RASR 3\X0D  |
| Aspect Ratio Full Input RGB 1         | RASR 5\X0D  |
| Aspect Ratio Normal Input RGB 1       | RASR 1\X0D  |
| Aspect Ratio Stretch Input RGB 1      | RASR 2\X0D  |
| Aspect Ratio V-Stretch Input RGB 1    | RASR 11\X0D |
| Aspect Ratio Area Zoom Input RGB 2    | RBSR 10\X0D |
| Aspect Ratio Border Input RGB 2       | RBSR 6\X0D  |
| Aspect Ratio Dot by Dot Input RGB 2   | RBSR 3\X0D  |
| Aspect Ratio Full Input RGB 2         | RBSR 5\X0D  |
| Aspect Ratio Normal Input RGB 2       | RBSR 1\X0D  |
| Aspect Ratio Stretch Input RGB 2      | RBSR 2\X0D  |
| Aspect Ratio V-Stretch Input RGB 2    | RBSR 11\X0D |
| Aspect Ratio Area Zoom Input S-Video  | RASV 10\X0D |
| Aspect Ratio Border Input S-Video     | RASV 6\X0D  |
| Aspect Ratio Dot by Dot Input S-Video | RASV 3\X0D  |
| Aspect Ratio Full Input S-Video       | RASV 5\X0D  |
| Aspect Ratio Normal Input S-Video     | RASV 1\X0D  |
| Aspect Ratio Stretch Input S-Video    | RASV 2\X0D  |
| Aspect Ratio V-Stretch Input S-Video  | RASV 11\X0D |
| Aspect Ratio Area Zoom Input Video    | RBSV 10\X0D |
| Aspect Ratio Border Input Video       | RBSV 6\X0D  |
| Aspect Ratio Dot by Dot Input Video   | RBSV 3\X0D  |
| Aspect Ratio Full Input Video         | RBSV 5\X0D  |
| Aspect Ratio Normal Input Video       | RBSV 1\X0D  |
| Aspect Ratio Stretch Input Video      | RBSV 2\X0D  |

## Global Scripter Module Communication Sheet

Revision: 1/19/2018

|                                           |             |
|-------------------------------------------|-------------|
| <b>Aspect Ratio V-Stretch Input Video</b> | RBSV 11\x0D |
| <b>Auto Image None</b>                    | ADJS 1\x0D  |
| <b>Freeze Off</b>                         | FREZ 0\x0D  |
| <b>Freeze On</b>                          | FREZ 1\x0D  |
| <b>Input DVI</b>                          | IRGB 3\x0D  |
| <b>Input RGB 1</b>                        | IRGB 1\x0D  |
| <b>Input RGB 2</b>                        | IRGB 2\x0D  |
| <b>Input S-Video</b>                      | IVED 1\x0D  |
| <b>Input Video</b>                        | IVED 2\x0D  |
| <b>Lamp Mode Eco</b>                      | THMD 1\x0D  |
| <b>Lamp Mode Normal</b>                   | THMD 0\x0D  |
| <b>Power On</b>                           | POWR 1\x0D  |
| <b>Power Off</b>                          | POWR 0\x0D  |
| <b>Volume 0</b>                           | VOLA 00\x0D |
| <b>Volume 60</b>                          | VOLA 60\x0D |

## Appendix B. Update Commands

|                                   |              |
|-----------------------------------|--------------|
| <b>AV Mute</b>                    | IMBK????\x0D |
| <b>Aspect Ratio Input DVI</b>     | RCSR????\x0D |
| <b>Aspect Ratio Input RGB 1</b>   | RASR????\x0D |
| <b>Aspect Ratio Input RGB 2</b>   | RBSR????\x0D |
| <b>Aspect Ratio Input S-Video</b> | RASV????\x0D |
| <b>Aspect Ratio Input Video</b>   | RBSV????\x0D |
| <b>Device Status</b>              | TABN 1\x0D   |
| <b>Freeze</b>                     | FREZ????\x0D |
| <b>Input</b>                      | ICLK????\x0D |
| <b>Lamp Mode</b>                  | THMD????\x0D |
| <b>Lamp Status</b>                | TLPS 1\x0D   |
| <b>Lamp Usage</b>                 | TLTT 1\x0D   |
| <b>Power</b>                      | POWR????\x0D |
| <b>Volume</b>                     | VOLA????\x0D |